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Secure On-Demand Dynamic Routing for Mobile Ad hoc Networks

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ABSTRACT
There square measure finding a route to destination, new route discovery is initiated. The frequent discoveries result in a lot of network congestion. To avoid this multipath routing protocols are planned to seek out multiple routes to destination and put on to alternate secondary path just in case of route broken and its offer higher routing performance and security. In this developed research paper attempt to compare the performance of two reactive routing protocols for MANETs that is Secure Dynamic Source Routing (SDSR) and Dynamic Source Routing (DSR), on-demand gateway discovery protocol where a mobile device of MANET gets connected to gateway. SDSR using the Intrusion Detection system (IDS) and trust based routing, the performance results are analysed by varying simulation time. Besides, every one of the previously mentioned conventions are thought about in light of a few significant presentation measurements which are bundle conveyance proportion, start to finish postponement and normal energy.

Keywords: *MANET, SDSR, route, destination, routing, protocols*

INTRODUCTION

The recent trends in wireless communication network have changed the lives of human beings. The new wireless technologies create a potential for the next generation MANET and applications. Wireless technologies such as Bluetooth or the 802.11 standards enable mobile devices to establish a MANET by connecting dynamically through the wireless medium without any centralized structure. MANET is a network having dynamic topology that consists of mobile nodes without base Station or centralized control. All mobile nodes carry out working of switches that pursuit and keep up with courses to different hubs in the organization. Trouble in impromptu organization is that for correspondence with different hubs, a hub should be in the transmission scope of base station yet in some cases a hub moves and organization fizzles. MANET has tackled this issue as in MANET hubs follow multi-jump design for speaking with different hubs. Directing is the most common way of getting data across web work from source to objective by choosing best cordial way that a parcel needs to take in internetwork. To play out this, a bunch of directing conventions required that utilizes measurements to track down ideal way for a parcel to travel. Directing conventions planning objectives are optimality, effortlessness, low above, strength, dependability and adaptability. Course solicitation and Course answer messages are utilized to find and store the ways found from the source to objective. Subsequent to finding the ways, briefest way is chosen by the source hub. Ways found by most brief way calculation cause issues like clog issues as the focal point of organization convey more traffic, this outcomes in terrible showing. The fundamental objectives of multipath directing conventions are to keep up with solid correspondence, to lessen steering above by utilization of auxiliary ways, to guarantee load adjusting, to work on nature of administration, to stay away from the extra course revelation above. The major focus of the routing is the performance and the efficiency of the protocol in the presence of a dynamic network environment. The routing protocol has to overcome the security pitfalls to utilize the potentials of the MANET. A secure routing is challenging due to the security vulnerabilities present in the active network.[1-5]

BACKGROUND WORK

This work evolves the previous work that has been undergone in this field. An execution investigation of AODV directing convention was finished by Das, et al (2000). Improvement and examination of multipath directing convention AOMDV in light of CMMBCR was finished by Yin-jun Yang (2011). A hub disjoint multipath directing strategy in view of AODV convention for MANET was finished by Lal, *et al*. (2012). Expanded throughput for load based diverts mindful directing in MANETs with reusable ways was finished by Ayyasamy (2012). Detail based interruption discovery for automated airplane frameworks was finished by Mitchell (2012). A study of assaults on MANET directing conventions was finished by Tayal (2013). Audit on MANET steering conventions and difficulties were finished by Habib (2013). An overview of interruption identification in remote organization applications was finished by Mitchell (2014). Conduct rule particular based interruption recognition for security basic clinical digital actual frameworks was finished by Mitchell (2015). Confided in secure specially appointed on-request multipath distance vector steering in MANET was examination by Abrar Omar Alkhamisi (2016).[6-11]

EXISTING METHOD

DSR (Dynamic Source Routing)

In DSR, at the point when a course is required from source to objective, then, at that point, source begins a course disclosure process by flooding a RREQ for objective. With the assistance of grouping numbers, RREQs are remarkably recognized so that copy RREQs can be distinguished and disposed of. At the point when a non-copy RREQ is gotten then middle of the road hub records past bounce and quest for a new course passage to the objective in directing table. On the off chance that new course is available, the hub sends a RREP to the source however in the event that new course is absent, it rebroadcasts the RREQ. The steering data is refreshed by a hub provided that a RREP contains either a bigger objective succession number than past one or a course with less jump count found. DSR routing process with the help of IDS and trust-based routing, the attack identification and isolation in DSR are carried out in two phases of routing such as route discovery phase and data forwarding phase. The trust obtained from IDS is applied in the routing decision-making about propagating RREQ packet of the source and selecting the trust based router for data forwarding. Trust based route discovery process initially, each node assigns the trust value as ‘one’ to its neighboring nodes. According to the routing activities of these nodes, the IDS measure the original trust value and inform the network layer. The route discovery process is carried out based on the trust value to isolate the flooding attacker activity. Prior to rebroadcasting the received RREQ packet to the neighbors, every node checks for the trust value of the source that has broadcasted the RREQ packet. If the trust value is lesser than the threshold, then the RREQ packet from the corresponding source is dropped to block the flooding activity of the attacker. For instance, in the trust value of the source node is maintained by its neighboring nodes such as A, B, and C. The trust value of the source node is different in various neighboring nodes due to the network collisions. The trust value of the source node is very less, so these nodes do not forward the RREQ packet of the source node into the network. Trust based data forwarding process data forwarding process is carried out based on the trust value to isolate the activity of black hole and the grayhole attacker. Prior to forwarding the data packet to the router, every node checks for the trust value of the router. The trust value of the router indicates the reliability of data delivery through it. If the trust value is low, the current data transmission through the malicious router is blocked. Subsequently, the trusted router from the routing table is accessed to resume data transmission through it. For instance, the trust value of the node B is higher than node A, and so the source node selects router B for further data forwarding. Thus, the IDS and the trust based DSR protocol improve the routing performance and security in MANET.[12,13]

PROPOSED METHOD

SDSR (Secure Dynamic Source Routing)

In this proposed work TS-DSR, at the point when a course is required from source to objective, then, at that point, source begins a course disclosure process by flooding a RREQ for objective. With the assistance of grouping numbers, RREQs are remarkably recognized so that copy RREQs can be distinguished and disposed of. At the point when a non-copy RREQ is gotten then middle of the road hub records past bounce and quest for a new course passage to the objective in directing table. On the off chance that new course is available, the hub sends a RREP to the source however in the event that new course is absent, it rebroadcasts the RREQ. The steering data is refreshed by a hub provided that a RREP contains either a bigger objective succession number than past one or a course with less jump count found. SDSR routing process with the help of IDS and trust-based routing, the attack identification and isolation in SDSR are carried out in two phases of routing such as route discovery phase and data forwarding phase. Trust Based Route Discovery Process Initially, each node assigns the trust value as 'one' to its neighboring nodes. Prior to rebroadcasting the received RREQ packet to the neighbors, every node checks for the trust value of the source that has broadcasted the RREQ packet. If the trust value is lesser than the threshold, then the RREQ packet from the corresponding source is dropped to block the flooding activity of the attacker. For instance, in the trust value of the source node is maintained by its neighboring nodes such as A, B, and C. The trust value of the source node is different in various neighboring nodes due to the network collisions. The trust value of the source node is very less, so these nodes do not forward the RREQ packet of the source node into the network. For instance, the trust value of the node B is higher than node A, and so the source node selects router B for further data forwarding. Thus, the IDS and the trust based MTS-DSR protocol improve the routing performance and security in MANET.[14,15]

EXPERIMENTAL SETUP

The NS-2 which is a discrete event driven simulator developed at UC Berkeley is used in this simulation process. Network Simulator NS-2 is useful in designing new protocols, comparison of different protocols and for evaluating the traffic. A scenario file is taken as input in the NS-2 simulation process that shows the motion of each node and the originating packets by each node. The simulation models are built using the NS-2 version 2.34 and it is run under bandwidth of 40 MHZ.[16]

Table 1: *Simulation parameters.*

Parameter	Values
Simulation area	680*680
Number of nodes	50
Number of packets sender	25
Constant bit rate	4 (packets/second)
Packet size	512 bytes
Initial energy/node	100 joules
Antenna model	Omni directional
Simulation time	500 sec

RESULTS & DISCUSSION

The results for the above mentioned simulation experiment is shown with the help of graphs. In this section the comparison of DSR and SDSR is made and the metrics used for the analysis of results are packet delivery ratio and end to end delay, energy.

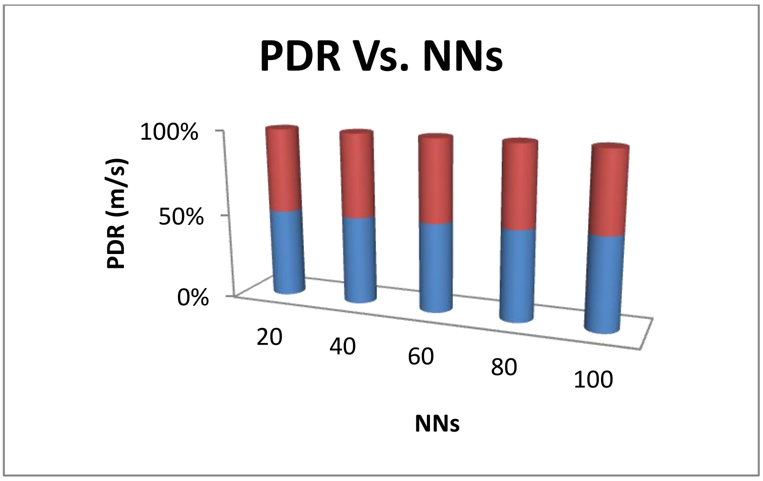

Fig. 1: *Packet delivery ratio Vs. number of nodes.*

It is observed from Fig 1that when compared with DSR algorithm, SDSR increases the delivery ratio above 85% with the increase in the number of nodes from 20 to 100. As the proposed algorithm finds maximum secure and lowest link failed route with minimum retransmit packets frequently, it is possible to increase the delivery ratio.

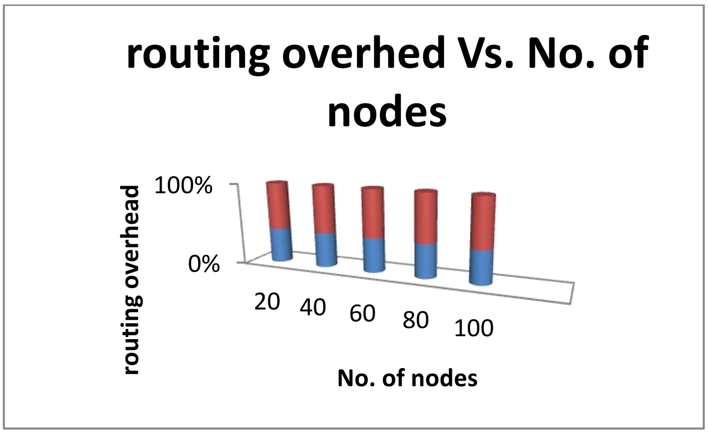

Fig. 2: *Routing overhead Vs. number of nodes.*

SDSR has the lowest overhead in comparison with DSR, when there are 680 of transmission ranges. When the transmission range increases, the connectivity among the nodes also increases, which enables the proposed method to identify more number of alternate secondary paths which in turn reduces the congestion. Fig 2 describes the decrease in overhead obtained by the proposed SDSR when there are 20 to 100 nodes.

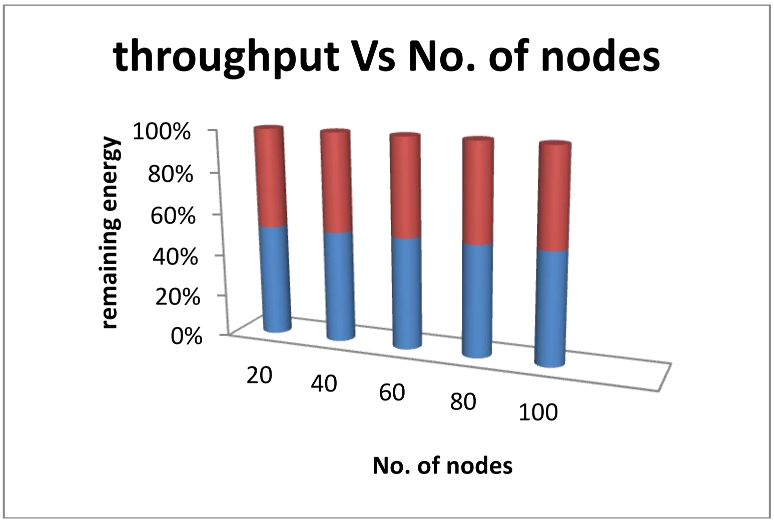

Fig. 3: *Throughput Vs. number of nodes.*

Fig 3 shows the graph of the throughput when the topology is size 680m, the number of nodes increased from 20 to 100. The proposed SDSR increases the throughput with the increasing topology size compared to DSR and SDSR.

CONCLUSION & FUTURE WORK

SDSR, the multipath enhancement to DSR was designed generally for highly dynamic ad hoc networks where link fails and route breaks occur frequently. The comparison was done on basis of packet delivery ratio, end-to-end delay, and energy.

On the basis of simulation results it is concluded that SDSR is better than DSR. SDSR outperforms DSR due its ability to discover alternate routes when a current link fails. In spite of the fact that DSR causes additional steering overheads due to flooding the organization and parcel defers due its backup course of action disclosure process, it is a lot of proficient in the event of bundle conveyance for a similar explanation. SDSR proves to be more efficient than DSR as it provides better throughput. Finally the conclusion is that when network load increases SDSR is a better on-demand routing protocol than DSR as it gives better statistics for packet delivery and throughput were implementing via network simulator 2.

To increase the merits of this research work, there is a plan to investigate the following issues in our future research. The same concept can be applied in satellite to increase delivery ratio and reduce delay in the route and also to save more energy. The performance of SDSR can be tested in real time network environment.

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BIOGRAPHICAL

Prof. K. Thamizhmaran received his M.E degree from Faculty of Engineering Annamalai University, Tamilnadu in the year 2012 respectively, M.sc in yoga degree from Faculty of Education, Annamalai University, Tamilnadu in the year 2019 respectively and Post Graduate in Guidance & Counselling degree from Faculty of Psychology, Annamalai University and Tamilnadu in the year 2020 respectively. He is currently pursing Doctor of Philosophy in Mobile Ad-hoc Network (Network Security), Faculty of Engineering Annamalai University from 2013 to till date.

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